

HabitFlow API Specification

Team-404

Version 1.1

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Prepared by: Renz Mathieu M. Raquin (Task API Spec)

Endpoints marked with an Authorization parameter require the header Authorization: Bearer <token> from /api/auth/login. Every authenticated request only sees data owned by that user.

III. Tasks (api/tasks)

Every task belongs to a project. Tasks created without a projectId are placed in the user's Inbox project.

Each task field has its own update endpoint (III.E to III.M) for quick edits, and Update Task (III.D) changes several basic details at once. isArchived is always false when a task is created and is changed only through Archive / Unarchive Task (III.N). Archived tasks appear only on the Archive page (List Tasks with archived=true) and can be restored or deleted from there.

A. List Tasks

API Endpoint	/api/tasks	
method	GET	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
projectId	integer	Optional query parameter. Returns only the tasks of this project. Omit for all tasks.
archived	boolean	Optional query parameter. true returns only archived tasks (Archive page), false only active ones. Default is false.
<i>Retrieves task records for the authenticated user, ordered by sortOrder. Archived tasks are only returned when archived=true.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
[]	array	List of tasks.
[], taskId	integer	Task ID.
[], projectId	integer	Project ID.
[], priorityId	integer	Priority ID.
[], title	string	Task title.
[], description	string/null	Task description.
[], dueDate	string/null	Due date (YYYY-MM-DD).
[], dueTime	string/null	Due time (HH:mm:ss).

[],deadline	string/null	Deadline (YYYY-MM-DD).
[],completedAt	string/null	Completion timestamp.
[],sortOrder	integer	Display order within the project.
[],isRecurring	boolean	True if the task has a recurrence rule.
[],isArchived	boolean	True if the task is in the archive.
[],labels	array	Labels attached to the task.
[],labels[],labelId	integer	Label ID.
[],labels[],name	string	Label name.
[],labels[],color	string/null	Label color (hex).
Sample Response		
[{"taskId": 1, "projectId": 10, "priorityId": 4, "title": "Study ILT Module 4", "description": "Read all lessons in the module and finish the code", "dueDate": "2026-10-02", "dueTime": "18:00:00", "deadline": "2026-10-02", "completedAt": null, "sortOrder": 0, "isRecurring": false, "isArchived": false, "labels": [{"labelId": 3, "name": "Study", "color": "#2563EB"}]}		
Fail		
Status Code	401	
Response Type	json	
Response Data	Type	Description
error	string	Authentication error.
Sample Response		
{"error": "Unauthorized"}		

B. Get Task

API Endpoint	/api/tasks/{taskId}	
method	GET	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
<i>Returns a single task owned by the authenticated user, active or archived.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.
projectId	integer	Project ID.
priorityId	integer	Priority ID.

title	string	Task title.
description	string/null	Task description.
dueDate	string/null	Due date (YYYY-MM-DD).
dueTime	string/null	Due time (HH:mm:ss).
deadline	string/null	Deadline (YYYY-MM-DD).
completedAt	string/null	Completion timestamp.
sortOrder	integer	Display order within the project.
isRecurring	boolean	True if the task has a recurrence rule.
isArchived	boolean	True if the task is in the archive.
labels	array	Labels attached to the task.
labels[].labelId	integer	Label ID.
labels[].name	string	Label name.
labels[].color	string/null	Label color (hex).
Sample Response		
<pre>{ "taskId": 1, "projectId": 10, "priorityId": 4, "title": "Study ILT Module 4", "description": "Read all lessons in the module and finish the code", "dueDate": "2026-10-02", "dueTime": "18:00:00", "deadline": "2026-10-02", "completedAt": null, "sortOrder": 0, "isRecurring": false, "isArchived": false, "labels": [{ "labelId": 3, "name": "Study", "color": "#2563EB" }] }</pre>		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task not found or not accessible.
Sample Response		
<pre>{ "error": "Task not found" }</pre>		

C. Add Task

API Endpoint	/api/tasks	
method	POST	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
projectId	integer	Optional. Existing project ID. Omit to add the task to the Inbox.
priorityId	integer	Priority ID (default is 4).
title	string	Task title (required, max 200 chars).
description	string/null	Optional description.

dueDate	string/null	Optional due date.
dueTime	string/null	Optional due time. Requires dueDate.
deadline	string/null	Optional deadline. Must not be earlier than dueDate.
sortOrder	integer	Display order (default is 0).
<i>Creates a new task. Tasks without a projectId are placed in the user's Inbox project. New tasks always start with isArchived set to false.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	New task ID.
projectId	integer	Project ID (the Inbox if none was given).
priorityId	integer	Assigned priority.
title	string	Task title.
description	string/null	Task description.
dueDate	string/null	Due date.
dueTime	string/null	Due time.
deadline	string/null	Deadline.
sortOrder	integer	Display order.
isArchived	boolean	Archived flag (always false).
Sample Response		
<pre>{ "taskId": 25, "projectId": 1, "priorityId": 2, "title": "Submit documentation", "description": "Upload the final document", "dueDate": "2026-10-02", "dueTime": "17:00:00", "deadline": "2026-10-02", "sortOrder": 0, "isArchived": false }</pre>		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
<pre>{ "error": "Task title is required" }</pre>		

D. Update Task

API Endpoint	/api/tasks/{taskId}	
method	PUT	
Request Parameters	Type	Description

Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
projectId	integer	Moves the task to another project owned by the user.
title	string	Updated title (max 200 chars).
description	string/null	Updated description.
priorityId	integer	Updated priority level (1 to 4).
<i>Updates several details of a task at once (e.g. from an Edit Task form). Fields not sent are left unchanged. Labels, due date, due time, deadline, sort order and archive status are changed only through their own endpoints (III.I to III.N). Use Reschedule Task (IV.C) to change several scheduling fields at once and Complete / Reopen Task (IV.D) for completion.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.
message	string	Success confirmation.
Sample Response		
{"taskId": 2, "message": "Task updated successfully"}		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
{"error": "Task title cannot be empty"}		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task or target project not found.
Sample Response		
{"error": "Invalid updating task."}		

E. Move Task to Project

API Endpoint	/api/tasks/{taskId}/project
method	PUT

Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
projectId	integer	Target project ID owned by the user (e.g. to move a task out of the Inbox).
<i>Moves a task to another project. The task keeps its other details.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.
projectId	integer	New project ID.
message	string	Success confirmation.
Sample Response		
{ "taskId": 25, "projectId": 10, "message": "Task moved successfully" }		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
{ "error": "projectId is required" }		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task or target project not found or not accessible.
Sample Response		
{ "error": "Project not found" }		

F. Rename Task

API Endpoint	/api/tasks/{taskId}/title	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.

taskId	integer	Existing task ID (path parameter).
title	string	New task title (required, max 200 chars).
<i>Changes only the title of a task.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.
title	string	Updated title.
message	string	Success confirmation.
Sample Response		
<pre>{"taskId": 25, "title": "Submit final documentation", "message": "Task renamed successfully"}</pre>		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
<pre>{"error": "Task title is required"}</pre>		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task not found or not accessible.
Sample Response		
<pre>{"error": "Task not found"}</pre>		

G. Update Task Description

API Endpoint	/api/tasks/{taskId}/description	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
description	string/null	New description. null clears it.
<i>Changes only the description of a task.</i>		

Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.
description	string/null	Updated description.
message	string	Success confirmation.
Sample Response		
{ "taskId": 25, "description": "Upload the signed PDF", "message": "Task description updated successfully" }		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task not found or not accessible.
Sample Response		
{ "error": "Task not found" }		

H. Update Task Priority

API Endpoint	/api/tasks/{taskId}/priority	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
priorityId	integer	New priority (1 is highest, 4 is lowest). See List Priorities (V.A).
<i>Changes only the priority of a task.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.
priorityId	integer	Updated priority.
message	string	Success confirmation.
Sample Response		
{ "taskId": 25, "priorityId": 1, "message": "Task priority updated successfully" }		

Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
{"error": "Invalid priority"}		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task not found or not accessible.
Sample Response		
{"error": "Task not found"}		

I. Set Task Labels

API Endpoint	/api/tasks/{taskId}/labels	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
labelIds	array of integer	Complete list of label IDs for the task. An empty array removes all labels.
<i>Replaces the labels attached to a task. Labels themselves are created and managed in Task Organization (V.B to V.E).</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.
labels	array	Labels now attached to the task.
labels[].labelId	integer	Label ID.
labels[].name	string	Label name.
labels[].color	string/null	Label color (hex).
message	string	Success confirmation.
Sample Response		

{"taskId": 25, "labels": [{"labelId": 3, "name": "Study", "color": "#2563EB"}, {"labelId": 4, "name": "Habit", "color": "#16A34A"}], "message": "Task labels updated successfully"}		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task or label not found or not accessible.
Sample Response		
{"error": "Invalid task or label."}		

J. Update Task Due Date

API Endpoint	/api/tasks/{taskId}/due-date	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
dueDate	string/null	New due date (YYYY-MM-DD). null clears the due date and the due time.
<i>Changes only the due date of a task. The deadline must not be earlier than the new due date. A recurring task's due date cannot be cleared.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.
dueDate	string/null	Updated due date.
dueTime	string/null	Due time (null if the due date was cleared).
status	string	Computed schedule status after the change (see IV.A).
message	string	Success confirmation.
Sample Response		
{"taskId": 25, "dueDate": "2026-10-05", "dueTime": "17:00:00", "status": "upcoming", "message": "Task due date updated successfully"}		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description

error	string	Validation error.
Sample Response		
{"error": "Deadline cannot be earlier than due date"}		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task not found or not accessible.
Sample Response		
{"error": "Task not found"}		

K. Update Task Due Time

API Endpoint	/api/tasks/{taskId}/due-time	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
dueTime	string/null	New due time (HH:mm:ss). null clears it. Requires the task to have a due date.
<i>Changes only the due time of a task.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.
dueTime	string/null	Updated due time.
status	string	Computed schedule status after the change (see IV.A).
message	string	Success confirmation.
Sample Response		
{"taskId": 25, "dueTime": "09:00:00", "status": "upcoming", "message": "Task due time updated successfully"}		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description

error	string	Validation error.
Sample Response		
{ "error": "Task must have a due date" }		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task not found or not accessible.
Sample Response		
{ "error": "Task not found" }		

L. Update Task Deadline

API Endpoint	/api/tasks/{taskId}/deadline	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
deadline	string/null	New deadline (YYYY-MM-DD). null clears it. Must not be earlier than dueDate.
<i>Changes only the deadline of a task.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.
deadline	string/null	Updated deadline.
message	string	Success confirmation.
Sample Response		
{ "taskId": 25, "deadline": "2026-10-07", "message": "Task deadline updated successfully" }		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
{ "error": "Deadline cannot be earlier than due date" }		

Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task not found or not accessible.
Sample Response		
{"error": "Task not found"}		

M. Update Task Sort Order

API Endpoint	/api/tasks/{taskId}/sort-order	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
sortOrder	integer	New display order within the project (min 0).
<i>Changes only the display position of a task in its project.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.
sortOrder	integer	Updated display order.
message	string	Success confirmation.
Sample Response		
{"taskId": 25, "sortOrder": 3, "message": "Task sort order updated successfully"}		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
{"error": "Sort order must be 0 or greater"}		
Fail		
Status Code	404	
Response Type	json	

Response Data	Type	Description
error	string	Task not found or not accessible.
Sample Response		
<code>{"error": "Task not found"}</code>		

N. Archive / Unarchive Task

API Endpoint	/api/tasks/{taskId}/archive	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
isArchived	boolean	true moves the task to the archive. false restores it from the archive.
<i>Archives a task or restores it from the archive. This is the only endpoint that changes isArchived. Archived tasks are hidden from List Tasks (unless archived=true), List Tasks by Schedule (IV.B) and the Task Dashboard counts (VI.A).</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.
isArchived	boolean	Updated archived flag.
message	string	Success confirmation.
Sample Response		
<code>{"taskId": 25, "isArchived": true, "message": "Task moved to archive"}</code>		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
<code>{"error": "isArchived is required"}</code>		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description

error	string	Task not found or not accessible.
Sample Response		
{"error": "Task not found"}		

O. Delete Task

API Endpoint	/api/tasks/{taskId}	
method	DELETE	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
<i>Soft deletes a task (sets deleted_at), whether it is active or archived (e.g. from the Archive page). Its recurrence rule and label links are removed.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
message	string	Task deletion confirmation.
Sample Response		
{"message": "Task deleted successfully"}		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task not found or not accessible.
Sample Response		
{"error": "Invalid deleting task."}		